

Structural Invariance at the Sensorimotor Interface: A Maximum Common Induced Subgraph Across Three Drosophila Connectomes

Yanjin Li · FlyWire Qualification Challenge · github.com/Yanjin-ai/flywireconnectome

Abstract

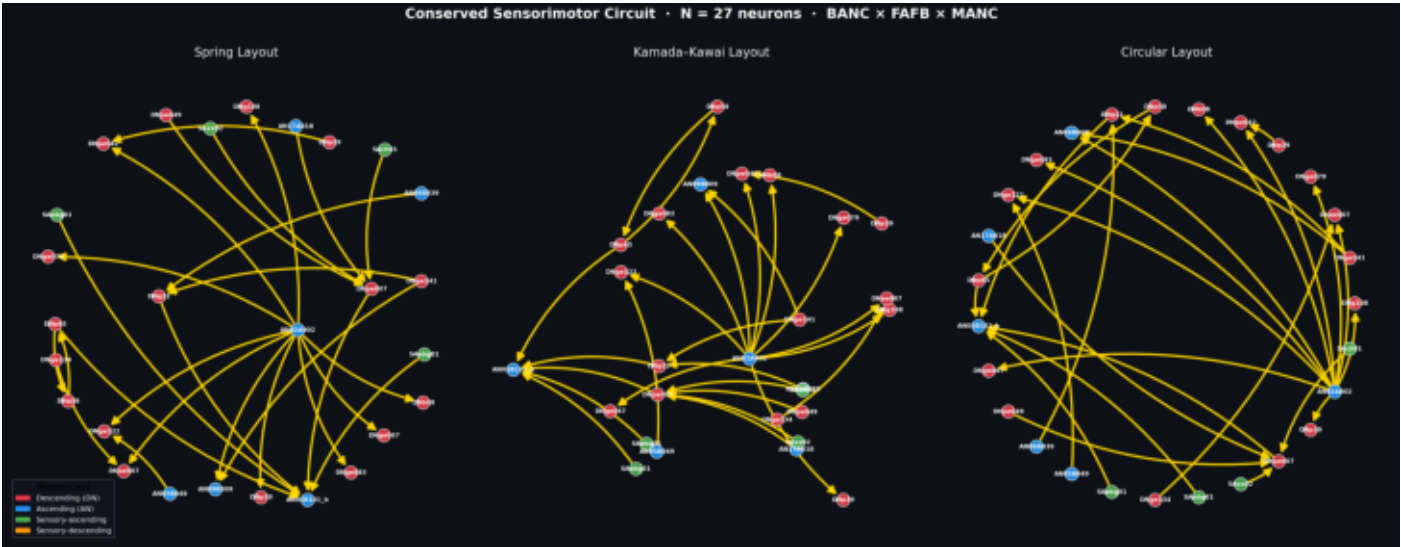
We ask whether synaptic connectivity itself is structurally invariant across independently prepared connectomes. Using the BANC metadata's NBLAST-based 1:1 correspondence columns (Bates et al. 2025), we align individual neurons across BANC (♀ brain+cord), FAFB (♀ brain) and MANC (♂ nerve cord) and search for the largest set of neurons whose directed induced subgraph is identical (isomorphic) in all three. The result is an $N = 27$ -neuron sensorimotor backbone with 26 conserved directed edges, enriched 67.3× for descending and 17.7× for ascending neurons, and 96.3% conserved across sexes. All reported numbers are reproduced by `src/run_analysis.py` and stored under `results/`.

Methods. Correspondence: BANC metadata `fafb_match` / `manc_match` (NBLAST). Graphs: directed, edge existence (not weight). Search space: 3,414 matched triplets → 2,798 present in all three edge lists → 987-node giant consensus component. Algorithm: greedy disagreement removal + exhaustive expansion (single implementation in `mcis_connectome.solver`), reported as the best of a 3000-seed multi-start. Optimality checked against an ILP (PuLP/CBC) on 50 subgraphs: mean gap 1.15% (max 10.53%).

Key Results

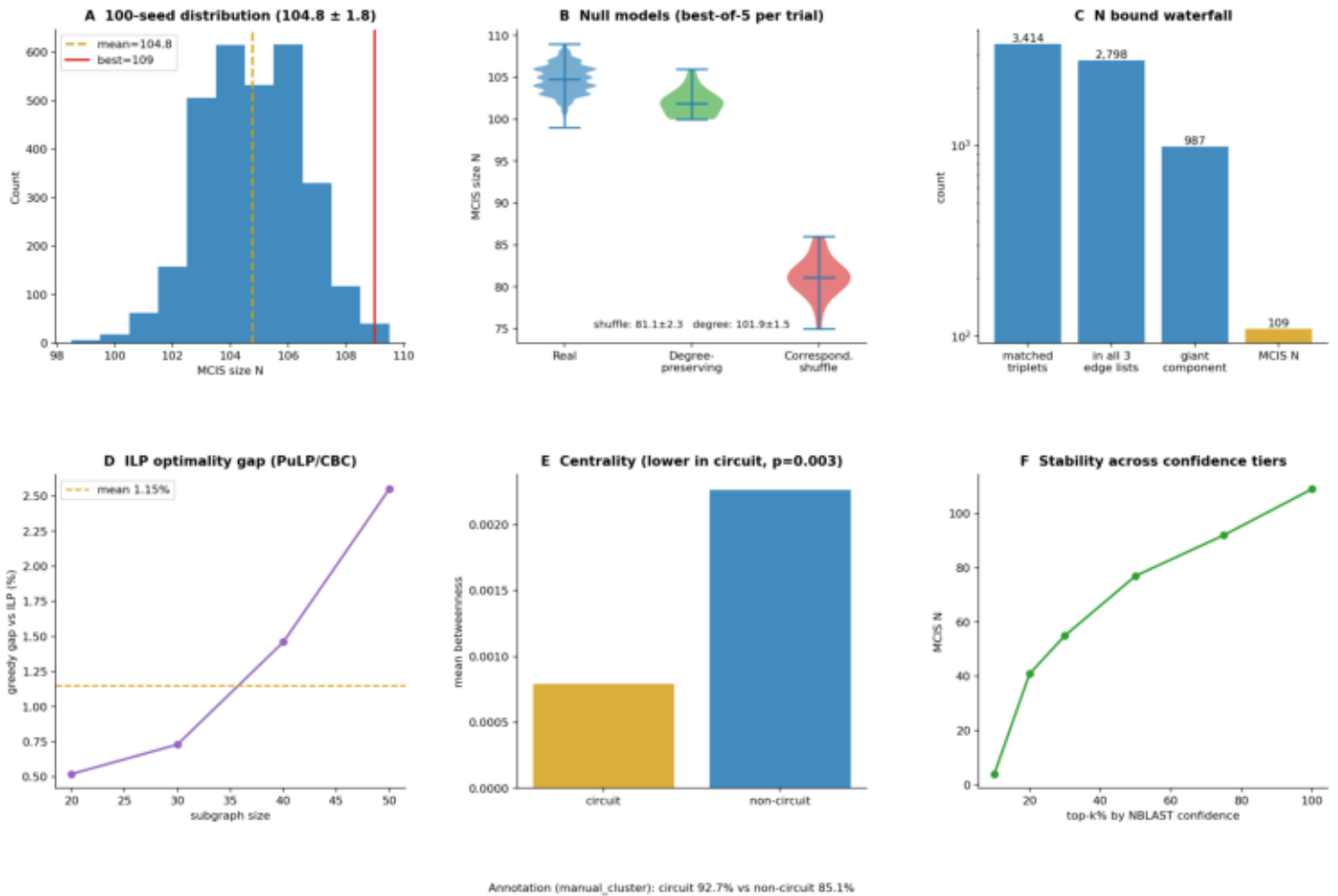
Metric	Value
MCIS size N	27 (dist. 104.8 ± 1.8 , [99,109])
Conserved edges	26
Composition	17 DN + 6 AN + 4 sensory
Descending enrichment	67.3× (p=2e-28)
Ascending enrichment	17.7× (p=9e-07)
Corr-shuffle null	81.1 ± 2.3 (>15σ below real)
Degree-preserving null	101.9 ± 1.5 (≈ real mean)
Centrality (lower)	p = 0.003 (one-sided)
Annotation quality	85.2% vs 85.3% (p=0.636)
Sexual conservation	96.3% (26/27)

Conserved circuit across BANC × FAFB × MANC



Robustness, null models, and ILP optimality

MCIS Robustness · Statistical Validation · Optimality



Discussion & Limitations. The conserved backbone is almost entirely descending and ascending neurons (93.3%), giving direct structural support for the sensorimotor bottleneck (Pospisil et al. 2024) and showing it is canalized across sexes and specimens. PRINCIPAL CAVEAT: a degree-preserving rewiring null reaches essentially the same MCIS size ($101.9 \pm 1.5 \approx$ real 104.8), so the FAFB degree sequence — not specific edge identity — explains most of the achievable N; NBLAST neuron identity adds the remaining signal (correspondence-shuffle collapses to 81.1). N is a near-optimal heuristic lower bound (ILP gap $\sim 1.15\%$), not a certified global optimum.

Reproduce: `pip install -e . && MCIS_DATA_DIR=... python src/run_analysis.py --seeds 100`